A compact learned Fourier correction for two-dimensional smoke simulation

Singularity

November 2026 · preprint

Abstract. A neural operator placed after a complete smoke solver can cost more than simply using a finer classical grid. We instead fold a learned Fourier correction with eight coefficients into the diffusion transform of a one-step transport solver with a midpoint trajectory. On 64 held-out synthetic smoke scenes the method ran at 0.889 ms per frame on one CPU and reduced mean density error by 8.8% relative to the best classical configuration within the same latency band (0.955 ms), with a paired interval of 4.6 to 13.1%; it improved 49 of 64 scenes and ran 55 times faster than the fine reference. The criterion was fixed before the test. The gain grows with the accuracy of the reference (9.8% against a 256×256 reference on eight predetermined scenes, 3.0% against a coarser 144×144 one), consistent with the correction recovering detail that coarse grids lose. The learned part adds only eight numbers to an existing solver.

1. Introduction

Semi-Lagrangian transport makes smoke simulation cheap enough for computer graphics [1]; a bounded MacCormack scheme improves accuracy at the cost of extra interpolation and limiting [2]. Fourier neural operators [3] learn field-to-field maps in frequency space and are a natural candidate for correcting a coarse solver. The practical question is not whether such a correction lowers error, but whether it does so at a cost below that of a finer classical grid. We study a design in which the learned correction reuses the Fourier transforms the solver already computes for diffusion, so it adds almost no time.

2. Methods

Smoke model. Buoyant smoke in a two-dimensional periodic square without obstacles. The state is density and vorticity; a divergence-free velocity is recovered through a Fourier stream-function solve, with a uniform lateral wind. A moving Gaussian source injects density and the horizontal density gradient drives vorticity. Viscosity and diffusion are Fourier multipliers. Scene seeds vary wind, source and initial vortices; one frame lasts 0.045 time units, and all outputs are compared on a 144×144 grid.

Transport and learned correction. The velocity at the middle of each interval is extrapolated as $\tilde{u}_j = 1.5u_j - 0.5u_{j-1}$ and the solver backtraces from x to $x - \Delta t\ \tilde{u}_j(x - \frac{1}{2}\Delta t\ \tilde{u}_j(x))$. After interpolation, source and forces, the same Fourier transforms used for diffusion apply the correction. For channel c and axis a, with $s_a = \sin^2(\pi k_a / n)$ and q_a the spatial mean of $4f_a(1-f_a)$, where f_a is the fractional departure displacement in cells,

$$M_c(k) = 1 + \operatorname{clip}\Big[\tfrac{1}{2} \sum_{a \in \{x,y\}} q_a\ (w_{c,a,1} s_a + w_{c,a,2} s_a^2),\ \ -0.18,\ 0.55\Big]. \tag{1}$$

The eight weights are fitted by regularised least squares against the next reference state on 48 training scenes. The filter is stored as two axis profiles; on development states it differs from the unfactored form by at most 3.8×10^{-6}.

Comparators and timing. Bounded MacCormack and linear or cubic interpolation with Euler, midpoint and temporal backtracing, over grid sizes from 48 to 144. The matched-cost comparator is the lowest-error classical method whose median step time is within 115% of the learned method. Warm timings on an AMD Ryzen 5 5625U include simulation and resampling to the common grid; method order is randomised per scene.

Study sequence. A first pilot (a two-layer Fourier network correcting a finished MacCormack step) and a second design (a spectral corrector on a single interpolated transport) were each tested on 24 new scenes and failed their criteria. The present design, with the 96×96 grid and filter strength chosen on 12 development scenes, then failed a first test of 64 scenes against a 144×144, four-substep reference, while an eight-scene check with a finer 192×192 reference favoured it. A new test was therefore fixed in writing before running: frozen model, 64 new scenes (32 with winds inside the training range, 32 above), primary reference 192×192 with eight substeps, the 144×144 reference reported in parallel, and a 256×256, 16-substep check on eight scenes chosen in advance. Primary criterion: at least 5% lower mean per-scene density RMSE than the matched-cost comparator, a positive lower bound of a paired stratified bootstrap interval, and at least a twofold speed-up over the fine reference. Secondary: at least 10% lower median time than the classical 112×112 temporal solver, with the interval on the relative error change below $+5\%$.

3. Results

Table 1. Held-out rollouts on 64 scenes of 40 frames; RMSE averaged per scene. The 108×108 row is the strongest classical method in the primary latency band.

method	grid	median ms	RMSE (ref. 192)	RMSE (ref. 144)
temporal linear, no filter	96^2	0.802	0.03115	0.02999
temporal linear (matched cost)	108^2	0.955	0.02779	0.02674
temporal linear	112^2	1.031	0.02690	0.02590
learned Fourier correction	96^2	**0.889**	**0.02535**	0.02594

Table 2. Dependence on the numerical reference. On the eight-scene check the mean errors are 0.03061 (learned) and 0.03394 (matched cost).

reference	error reduction vs matched cost	paired 95% interval	scenes improved
192×192, 8 substeps (primary)	**8.8%**	4.6 to 13.1%	49 / 64
144×144, 4 substeps	3.0%	-1.2 to 7.3%	39 / 64
256×256, 16 substeps (8 scenes)	9.8%	–	–



Figure 1. One frame of the first held-out scene on a common density scale. Left: fine reference. Middle: strongest classical solver at comparable latency. Right: learned correction.

- **Primary criterion met.** 8.8% lower error than the best classical solver at comparable cost, with a 55-fold speed-up over the fine reference (Table 1).
- **The filter matters.** On the same 96×96 grid and trajectory, the learned method has 18.6% lower error than the solver with its filter disabled, so the gain is not only the better trajectory.
- **Secondary criterion met.** Against the classical 112×112 solver the method saves 13.8% of median time while its error is 5.8% lower (upper interval endpoint -1.6%).
- **Larger against finer references.** 9.8% against the 256×256 check, 3.0% against the coarser 144×144 reference, where the interval includes zero (Table 2).

4. Discussion

The advance comes from combining a better departure trajectory with a learned correction that reuses an existing transform, so the correction is almost free in time. The finer the reference, the clearer the advantage, which is consistent with the correction recovering detail that a coarse reference itself lacks. Because the filter lives inside transforms that diffusion already needs, the same idea can be tried in any spectral smoke solver, with three-dimensional scenes and obstacles as the next steps.
Limitations. Synthetic periodic smoke without obstacles, one CPU and 40-frame sequences. The finer primary reference was adopted after a first test against the coarser one failed; the confirming test used new scenes and a frozen model.

Reproducibility. `experiment.py` runs, in order, development, the first test, the test with the finer reference and the figure; the two earlier designs are kept in `code/` (top level and `retry/`) with their own scripts. `results.json` contains all five stages: the two earlier designs, the first test, the confirmation and the 256×256 check. Scene seeds, weights, per-scene errors and frame timings are included.

Origin of the question. The question came from an automated forecast (Singularity v1, registered on 30 October 2026 with a SHA-256 fingerprint) that neural operators would enter computer graphics. The forecast concerned the arrival of the method, not the result of this study.

References

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